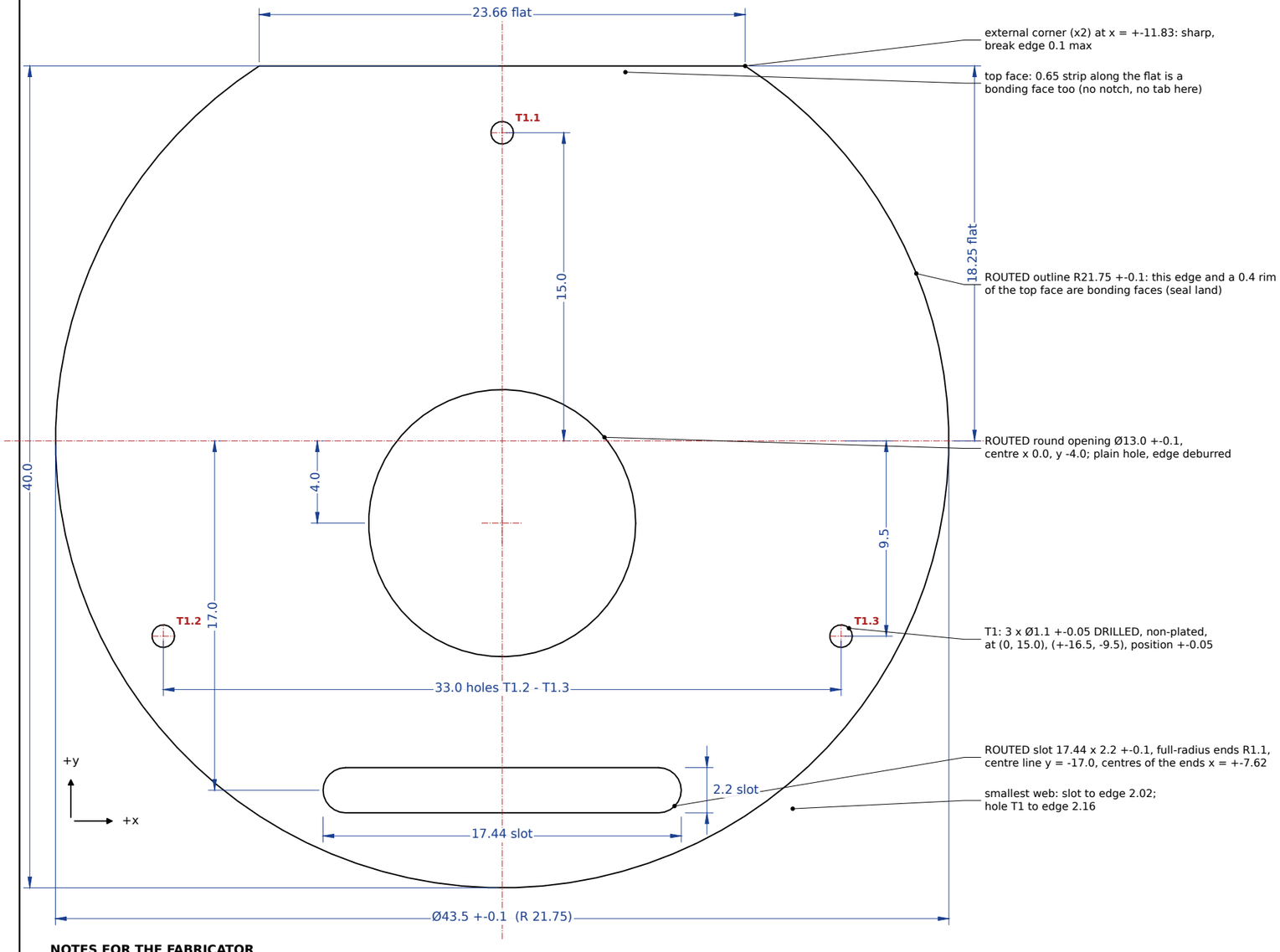
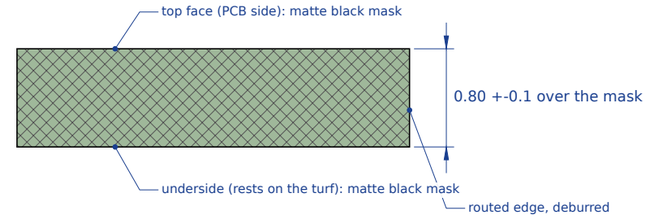
VIEW ON THE TOP FACE (PCB side), z = 0.80 4:1 datum: centre of the R 21.75 arc, +y toward the flat



EDGE VIEW (section y = 0 at the +x edge) 20:1

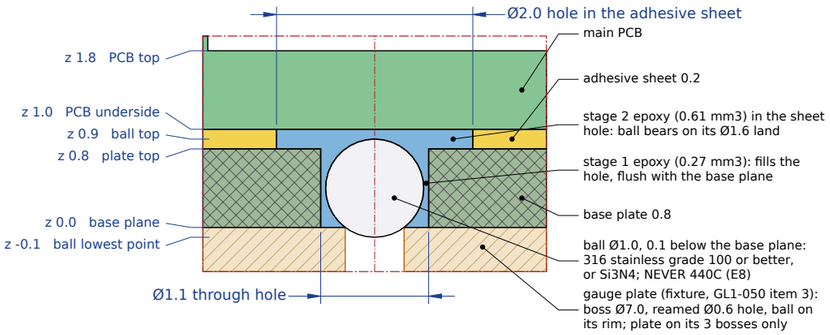


Tool / feature	Qty	Size	Tolerance	Position (x, y)	Plating
T1 drill	3	Ø1.1	+-0.05 dia, +-0.05 pos.	(0, 15.0), (+-16.5, -9.5)	none (NPTH)
Rout: round opening	1	Ø13.0	+-0.1	(0.0, -4.0)	-
Rout: slot	1	17.44 x 2.2, ends R1.1	+-0.1	centre (0, -17.0)	-
Rout: outline	1	R21.75, flat at y = 18.25	+-0.1	arc centre (0, 0)	-
Mask relief at T1	3 + 3	Ø1.25 opening, both faces	+-0.05	on the T1 holes	-
Thickness over mask	-	0.8	+-0.1	-	no copper

Web / edge distance (measured on the model)	mm
hole T1.1 to the outline	2.7
hole T1.2 to the outline	2.16
hole T1.3 to the outline	2.16
round opening to the outline	11.25
slot to the outline	2.02
slot to the round opening	5.4

Smallest internal radius: R1.1 (slot ends). No internal corner below R0.5; no drilled relief needed. A router bit of Ø2.0 or smaller makes every feature. Smallest web 2.02 mm.

FOR INFORMATION (not for fabrication): hole T1.1 in use, on the gauge plate 16:1



NOTES FOR THE FABRICATOR

1. MATERIAL: FR-4 glass-epoxy laminate, NO COPPER on any layer (unclad or fully etched): a mechanical plate, not a circuit. FINISHED THICKNESS OVER THE SOLDER MASK 0.8 +-0.1 mm. Choose the laminate for that finished thickness and state the build-up in the quote (a 0.8 mm copper-clad laminate ends near 0.75 without its copper). We use plates from 0.76 to 0.90 mm.
2. SOLDER MASK: matte black, both faces, full coverage EXCEPT a round relief Ø1.25 at each hole T1 on both faces: no mask on the hole wall or rim (or drill after mask). LEGEND on the underside only: sheet 2. No date code.
3. OUTLINE AND CUT-OUTS: routed, +-0.1. No V-score. Supply individually routed pieces. If panelised: break-off tabs ONLY on the flat (y = 18.25), mouse bites recessed into the outline by at most 0.25 mm, no nub proud of the outline anywhere. NO tab on the round part: it drops into a bore with 0.25 mm radial clearance and carries a 0.4 mm seal land.
4. HOLES T1: 3 x Ø1.1 +-0.05, drilled, non-plated, free of mask; position +-0.05 to each other, +-0.1 to the outline. They take Ø1.0 balls of 316 stainless (grade 100 or better) or silicon nitride, never 440C (E8; detail, for information).
5. CORNERS: slot ends full radius R1.1; no internal corner below R0.5. The two external corners at the ends of the flat are sharp (break edge 0.1 max).
6. FLATNESS: bow and twist 0.5 % maximum (0.2 mm over the part). Edges free of burrs and loose fibres. Both faces are bonding faces: no release agent, mask fully cured, parts clean.
7. DATA (mm, same origin): drawings/fab/GL1-011_base_plate_FR4-Profile.gbr (Gerber: outline + routed cut-outs), F_Mask.gbr / B_Mask.gbr (mask), B_Legend.gbr (legend, sheet 2), NPTH.drl (Excellon, NPTH); drawings/dxf/GL1-011_base_plate_FR4_fab.dxf (layers OUTLINE_ROUT, ROUT_OPENING_D13p0, ROUT_SLOT_17p44x2p2, DRILL_NPTH_D1p10); export/step/base_plate.step.
8. Datum: origin = centre of the R21.75 arc; +x to the right and +y toward the flat in this view (seen on the top face). The part is symmetric about the y axis.
9. To report with the first articles: thickness over the mask, hole diameters, outline. PROVISIONAL - not released. Nominal CAD dimensions (design/master_params.yaml rev D); checked by check 15 of the build.

Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

TITLE

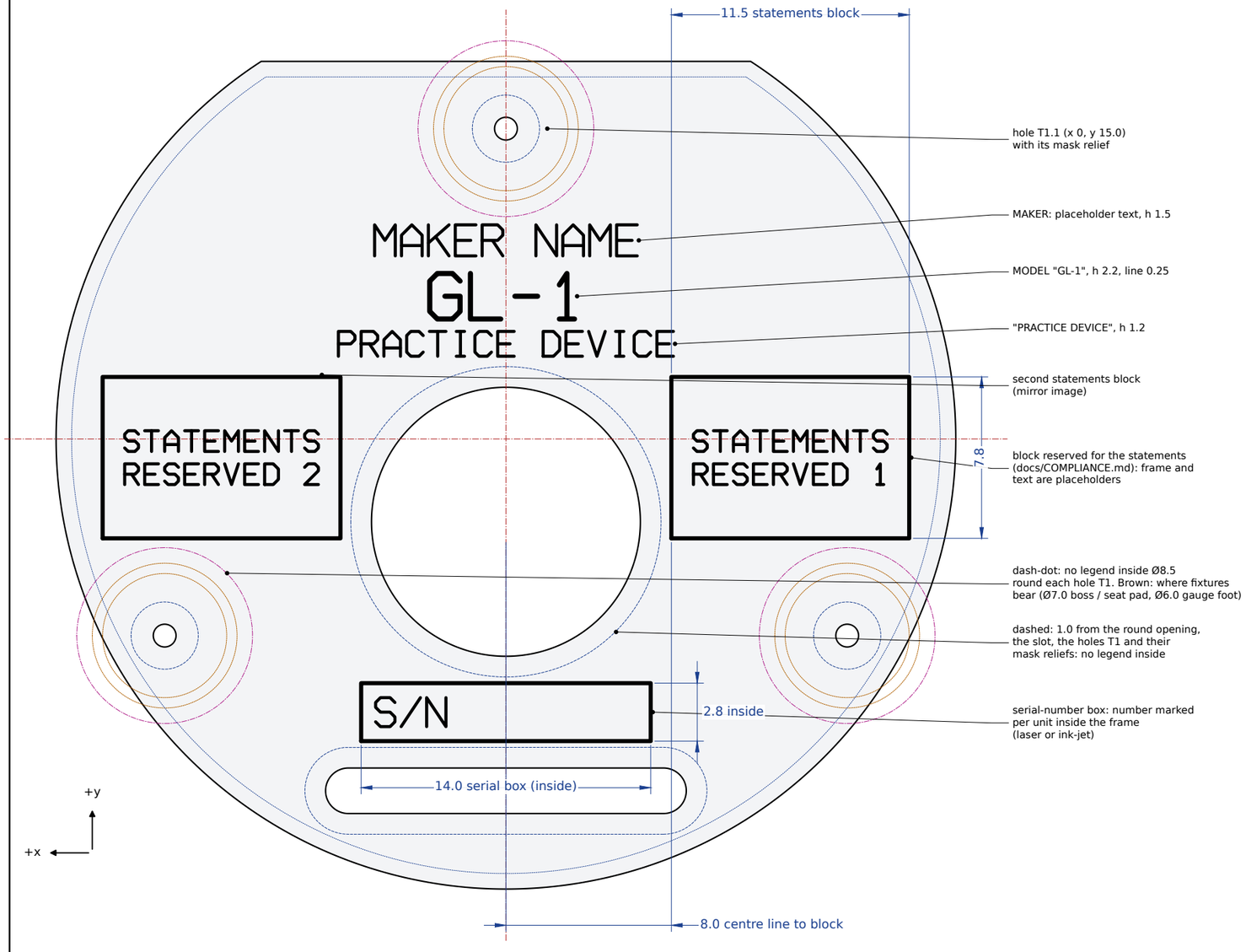
GL1-011 Base plate, FR-4 - FABRICATION DRAWING

PROJECT / PART

GreenLine GL-1 electronic ball marker - base plate (primary build)

MATERIAL / FINISH	PROCESS	
FR-4 0.80 mm, matte black solder mask both sides, NO COPPER	PCB fabricator: routed outline + cut-outs, drilled NPTH	
GENERAL TOLERANCE	SCALE (A3) / UNITS	
routed +-0.1, drilled +-0.05, thickness +-0.1	4:1 / 20:1 mm	
STATUS	DWG REV / PARAMS	SHEET / DATE
PROVISIONAL - not released	D-EVT1.2 / D-EVT1	1 of 2 2026-11-13

VIEW ON THE OUTER FACE (underside, as the legend is READ): seen from below, visor flat at the top 4:1 (+x of the part is to the LEFT)



Block	Drawn (placeholder)	u0, v0 .. u1, v1 as read	Text h / line	Room at text h 1.0
maker	MAKER NAME	-6.43, 8.78 .. 6.43, 10.43	1.5 / 0.15	-
model	GL-1	-3.81, 5.68 .. 3.44, 8.13	2.2 / 0.25	-
practice	PRACTICE DEVICE	-8.18, 3.93 .. 8.18, 5.28	1.2 / 0.15	-
statements_1	frame + STATEMENTS / RESERVED 1	7.9, -4.9 .. 19.6, 3.1	1.0 / 0.2	4 lines x 11 char.
statements_2	frame + STATEMENTS / RESERVED 2	-19.6, -4.9 .. -7.9, 3.1	1.0 / 0.2	4 lines x 11 char.
serial	frame + S/N	-7.1, -14.7 .. 7.1, -11.7	1.4 / 0.2	1 lines x 13 char.

Legend ink to (as drawn, computed)	mm	required
ball-foot hole 1 with its mask relief	3.97	>= 1.0 (YAML)
ball-foot hole 2 with its mask relief	3.98	>= 1.0 (YAML)
ball-foot hole 3 with its mask relief	3.98	>= 1.0 (YAML)
grain opening	1.2	>= 1.0 (YAML)
pad slot	1.2	>= 1.0 (YAML)
routed outline	1.57	>= 0.75
seat zones Ø8.5 round the holes T1	0.34	> 0 (outside)
fixture footprints with their play (gauge-plate boss, seat pad, bridge foot)	0.49	>= 0.15
space between letters at text h 1.0 (pitch 0.967)	0.15	>= 0.15

NOTES - LEGEND

1. LEGEND ON THE UNDERSIDE ONLY (the face that rests on the turf; 'bottom' in the data: GL1-011_base_plate_FR4-B_Legend.gbr). Like every bottom layer the file is given as seen from the top face: the text is mirrored in the file and reads correctly on the part, as drawn here. No legend on the top (PCB) face.
2. PROCESS: white legend print on the black mask, cured; or laser marking of the same artwork. Line width 0.15 or more, text height 1.0 or more, clear space between letters and between any two pieces of ink 0.15 or more (to be confirmed with the quote). Print thickness 0.02 typical, 0.03 maximum.
3. KEEP CLEAR: no legend within 1.0 mm of the holes T1 and their mask reliefs, the round opening and the slot (dashed), within 0.75 of the outline (dotted), or inside Ø8.5 round each hole T1 (dash-dot). Three fixtures bear on BARE MASK there (brown): the bosses of the ball gauge plate, the seat pads of GL1-050 and the feet of the bridge gauge that takes every depth reading on this face. Nothing may rest on the legend. Check 15 measures all of it on the legend layer read back.
4. PLACEHOLDERS: 'MAKER NAME', the two STATEMENTS frames and their labels stand for text that is not frozen (docs/COMPLIANCE.md rows 5 and 33). The final text goes inside the same frames; the frames themselves are then not printed. Room: about 88 characters at 1.0 mm in the two blocks (4 lines of 11 each, pitch 0.967): the short FCC sentence fits, the full two-condition statement does not (compliance checklist).
5. SERIAL-NUMBER BOX: frame 14.0 x 2.8 mm inside, line 0.2; 'S/N' printed. The number itself is marked per unit by us inside the frame, after test gate 1 (assembly step 13): laser preferred; an ink mark 0.03 thick at most. The box lies outside the seat zones.
6. PA12 alternative plate GL1-012: the same content and positions, laser-marked.
7. PROVISIONAL - not released. Nothing has been built.

Generated from the CAD model by cad/drawings.py - dimensions are nominal model values, computed, not typed. Frame: origin at puck centre on the base plane, +x right, +y toward the visor, +z up.

TITLE
GL1-011 Base plate, FR-4 - LEGEND of the outer face

PROJECT / PART
GreenLine GL-1 electronic ball marker - base plate (primary build)

MATERIAL / FINISH	PROCESS	
white legend print on the black mask, or laser mark	PCB fabricator: legend on the UNDERSIDE only	
GENERAL TOLERANCE	SCALE (A3) / UNITS	
legend to outline +-0.15; line >= 0.15, text >= 1.0	4:1 mm	
STATUS	DWG REV / PARAMS	SHEET / DATE
PROVISIONAL - not released	D-EVT1.2 / D-EVT1	2 of 2 2026-11-13